

Specification of Coursework-1

An Exercise in Designing a Software-Based System: *From requirements through to design-level models and scenario test cases*

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THIS IS AN INDIVIDUAL PROJECT

While discussion with fellow students as to the general nature of this project is acceptable, it is critically important that the solution you adopt and report are completely your own work. The re-use of other peoples work is not permitted and if identified will be treated as a disciplinary matter. Information on plagiarism can be found via

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1 Introduction

F28SD is assessed 100% by coursework. The coursework is divided into 2 parts, this document specifies what is required for the first part – Coursework-1.

2 The tasks involved in Coursework-1

You are required to develop a series of design models for a software-based system that supports the management of a *secure room*. The system is to be called *Secure Room Controller* (SRC). The remit for SRC is described informally below:

SRC shall manage access to a secure room. Access to the secure room is by means of an *air lock door* (ALD). The ALD has outer and inner doors. It is designed so that normally only one person can enter or leave a room at a time. To enter or leave a room managed by SRC, a person shall be required select an *access mode* using a touchscreen, i.e., one external touchscreen for entering the room and one internal touchscreen for exiting the room. SRC shall recognize two access modes, i.e., *staff-access* and *security-access*. If staff-access mode is selected by a person then they are required to supply a *staff identifier* (SID) and its associated *personal identification number* (PIN). The SID is a 6-character code while the PIN is a 4-digit code. Only when both the SID and PIN have been entered and validated is access to the room granted. The same protocol is required for entering as well as exiting the room. Meetings held within a room managed by SRC will be associated with a *security level* (SL), i.e., *top secret*, *secret* or *official*. All staff members have a *security clearance level*, i.e., SCL1, SCL2, SCL3. SCL1 is required for top secret meetings, SCL1 and SCL2 are required for secret meetings while staff with any security clearance level can attend an official meeting. A person's security clearance level can be changed at anytime. A new staff member is classified as *probationer* during the first 2 years of their employment. For top secret and secret meetings, a probationer's *line manager* must be present in the meeting room at all times. For each meeting, SRC is provided with a *meeting schedule* (MS) record. The MS will detail a planned start and end time for the meeting. The MS will also provide a list of participants, i.e., SIDs. Meetings can be scheduled within the room daily between the hours of 06:00 and 22:00. Daily at 02:00, SRC shall connect to an external *Global Meeting Scheduler* (GMS) system. The SRC will request from the GMS a collection of MS records for its associated room for the next 24-hours. The SRC shall be responsible for ensuring that people it allows into a room are consistent with the information held within the associated MS. SRC has the capability to activate a silent alarm that is monitored by the security guards within the building. Once the silent alarm is activated, no staff are allowed to entry the room. Moreover, anyone that attempts to exit the room will be held within the air lock, i.e., once the staff member enters the air lock then the outer and inner doors will be remain closed. Only a security guard is then able to open the outer door. To do this the *security-mode* needs to be selected via the external touchscreen, and then a 7-character security code that is unique to the SRC needs to be entered. During the course of a 24-hour period, SRC will create a log at the start of each meeting. This *meeting log* (ML) will record who actually attended the meeting, who failed to attend as well as anyone that was rejected when they attempted to enter. The exact start and end times will also be recorded in the ML. Each meeting will have its own ML. Each meeting will also have a *meeting chair* (MC). The start time for a meeting is defined to be when the MC enters the room. The MC must remain within the room during the duration of the meeting. Other participants are allowed to enter and leave multiple times during the meeting as long none of the constraints detailed above are violated. All entries and exits during the meeting period must be recorded within the ML as well as any security violations. An external *Human Resources System* (HRS) holds details of all staff members and can provide these details on demand, with the exception of PIN information. For example, if x denotes a valid SID, then `retrieveSCL( $x$ )` returns the SCL for staff member x , i.e., SCL1, SCL2 or SCL3. Crucially, SRC shall use HRS to

validate the staff access codes that it receives via the touchscreens. That is, when SRC supplies HRS with both a SID and PIN, HRS will respond to SRC by confirming whether or not the two codes are valid, i.e., SID is a valid staff ID and the PIN provided is correct with respect to the given SID. HRS can also provide information on probationers, i.e., whether or not a staff member is a probationer and the SID of their line manager. At 01:59 each day the SRC archives its ML records that were created during the previous 24-hours. This archive process shall involve SRC sending the ML records to an external *Meeting Archive System* (MAS). Once MAS has acknowledged receiving the records, SRC shall delete its own copy. A conventional emergency exit exists within a meeting room. Its use triggers an alarm. The emergency exit is completely independent of the SRC.

Based upon the informal description above, you are first required to define and model a set of requirements for SRC. Secondly, you are required to develop a set of design models that are consistent with your requirements. Thirdly, you are required to derive a set of scenarios test cases using aspects of your design models. **Guidance:** SRC needs to robust, e.g., it needs to be able to deal with erroneous and/or inconsistent inputs. Your requirements and design models should reflect this. The assessment of Coursework-1 will take into consideration the robustness of your design.

Specific Tasks:

The following tasks define the core work associated with Coursework-1. **Guidance:** while the tasks are describe as a sequence, you are recommended not to undertaken them purely sequentially. For example, the development of a Use Case specification will typically make you rethink your assumptions. Likewise, the development of a Sequence Diagram may make you rethink the structure of your Class Diagram. So think in terms of co-routining between tasks. Be prepared to revise and refine your decisions during the modelling process.

T1: The design remit above is incomplete, i.e., it contains **gaps** that need to be addressed in order for you to develop a design for SRC. Using a table, state the gaps that you identify within the design remit. Moreover, for each gap you should formulate an **assumption** that describes how the gap is to be addressed in your solution to the design problem. **Guidance:** you are not being asked to restate aspects of the above design remit. You are being asked to make explicit any gaps that you have identified in the design remit that are relevant to your design of SRC, and importantly how you have dealt with them, i.e., what assumptions you have made about SRC and its operational environment. Your assumptions should be realistic. For example, it is *not* realistic to expect that person will never supply erroneous data.

T2: Develop a set of **functional** and **non-functional**¹ **requirements**. **Guidance:** your requirements need to be sufficiently precise and complete that a third party, e.g., a fellow student, would be able to produce a design for SRC solely on the basis of your stated requirements, i.e., without sight of the informal design remit. Ensure that your requirements capture how bad behaviours should be dealt with as well as good behaviours. Moreover, for all requirements you should adopt a simple and uniform style – see the lecture notes entitled “*Requirements Engineering*” (see Week 1 on Canvas).

T3: Develop a **Use Case** model for SRC that includes at least the following 4 base use cases:

- *Retrieve Schedules*: describes how meeting schedules for a 24-hour period are set-up.
- *Enter Staff*: describes how a staff member enters a SRC managed meeting room.
- *Exit Staff*: describes how a staff member exits a SRC managed meeting room.
- *Archive Logs*: describes how meeting logs for a 24-hour period are archived.

¹Two non-functional requirements will be sufficient.

Specifically, your **Use Case** model should include **ONE Use Case Diagram** for SRC. Note that these base use cases may depend upon auxiliary use cases (i.e., inclusion/extension use cases). These auxiliary use cases should be shown in your **Use Case Diagram**. In terms of **Use Case Specifications** (textual descriptions), you should include specifications for the following 2 base use cases:

- *Enter Staff*: describes how a staff member enters a SRC managed meeting room.
- *Exit Staff*: describes how a staff member exits a SRC managed meeting room.

Note that you should include specifications for any auxiliary use cases that these two base use cases rely upon (i.e., inclusion/extension use cases). **Guidance:** don't forget to specify alternative flows. Make sure your diagram is consistent with the specifications that you provide.

T4: Provide a **traceability matrix** that links all your **Use Cases** with your **functional requirements**. **Guidance:** see lecture notes on “*UML Use Case Modelling (Part 1)*” (see Week 2 on Canvas).

T5: Develop a **Class Diagram** for SRC – include all the classes that will be part of the SRC software, and ensure that you include their associations. **Guidance:** GMS, HRS and MAS are external to SRC. Include their dependencies with SRC in your diagram but ensure that the system boundary for SRC is clearly shown. Ensure also that your **Class Diagram** is consistent with your **Sequence Diagram(s)**.

T6: Develop a **Sequence Diagram(s)** based upon your **Use Case Specification(s)** for *Enter Staff* (T3). **Guidance:** ensure that your model captures the alternative flows. Note that an alternative flow can be represented as a separate **Sequence Diagram**, or using conditional blocks within a single diagram, i.e., referred to as *interaction frames* in the lecture entitled “*UML Sequence Diagrams*” (see Week 4 on Canvas). Ensure that your **Sequence Diagram(s)** are consistent with your **Class Diagram**.

T7: Develop an **Activity Diagram** based upon your **Use Case Specification(s)** for *Exit Staff* (T3). **Guidance:** ensure that your model captures the alternative flows.

T8: Develop a **State Machine Diagram** that models the 24-hour cycle that SRC manages, i.e., starting from when SRC requests a MS. **Guidance:** ensure all transitions represent actions/events, e.g., a staff member supplies their SID. Ensure also that states represent key moments, e.g., meeting room occupied. Think in terms of the system as a whole. Consider for example a microwave, e.g., from its initial idle state, it can transition into a “ready-to-microwave” state when a user selects power and timing values. The event of the user pressing the ON button will then transition the microwave into its “microwaving” state. While the timer will normally trigger the next transition, a microwave usually also allows to user interrupt the microwaving state. All possible transitions need to be captured.

T9: Derive a set of **Scenario Test Cases** from the **Activity Diagram** you developed for a person exiting a meeting room, i.e., *Exit Staff*. For each **Scenario Test Case** clearly show the path(s) that it exercises within your **Activity Diagram** (T7). **Guidance:** for background on **Scenario Test Cases**, see the lecture notes entitled “*Software Verification: A Life-Cycle Perspective*” (see Week 7 on Canvas).

You **should** use the specification templates (i.e., Use Case templates) given on the Canvas course pages (see the “*UML Related Material*” module). UMLet is recommended for developing your UML diagrams, however, you may use an alternative drawing tool if you wish. But please note that hand-drawn UML diagrams are **not acceptable**.

3 The structure and content of your submission for Coursework-1

Your submission **should** take the form of a self-contained report (**MS Word or PDF format**)². Your report **should** contain page and section numbers, as well as a table of contents. Your report **should** include a **front page** that contains at least your name, student number, email address and your campus location. The **main body** of your report should contain 10 sections corresponding to the following 10 deliverables:

- D1:** A table that describes the **gaps** that you have identified within the given design remit. Each row of the table should describe a **gap** (first column) and the **assumption** (second column) you have made in order to address the **gap**. Clearly number each **gap** that you have identified. **[T1] (10-marks)**
- D2:** Tables containing your **functional** and **non-functional** requirements, again using a clear numbering scheme. **[T2] (12-marks)**
- D3:** Your **Use Case Diagram** for SRC. **[T3] (10-marks)**
- D4:** **Specifications** (textual descriptions) for the *Enter Staff* and *Exit Staff* base use cases within your **Use Case Diagram**, including any auxiliary use cases that they depend upon. **[T3] (10-marks)**
- D5:** Your **traceability matrix**. **[T4] (4-marks)**
- D6:** A **Class Diagram** for your SRC design. **[T5] (10-marks)**
- D7:** The **Sequence Diagram(s)** you derived from your *Enter Staff Use Case Specification(s)*. **[T6] (10-marks)**
- D8:** The **Activity Diagram** you derived from your *Exit Staff Use Case Specification(s)* **[T7] (10-marks)**
- D9:** Your **State Machine Diagram**. **[T8] (10-marks)**
- D10:** Your set of **Scenario Test Cases**. **[T9] (10-marks)**

Note that you are required to complete a Declaration of Authorship to confirm that all work you have submitted for individual assessment is your own (see 28SD Canvas section “Declaration of authorship AY24-25”).

4 Submission deadline for Coursework-1

Coursework-1 counts for 80% of the overall mark for the course. Your report should be submitted via Canvas. **4-marks** will be allocated to the quality of your report. The submission link can be found within the “Assessment” link of the F28SD “Course Navigation” panel. The **deadline** for submitting **Coursework-1** is **Monday 10 March 2025 (Week 9)**:

15:30 Edinburgh (local time)

23:59 Malaysia (local time)

17:00 Dubai (local time)

20:00 Aktobe (local time)

The standard penalty for the late submission of coursework will be applied unless evidence of Mitigating Circumstances is provided (see Undergraduate Programme Handbook for details).

²Note that MS Word allows you to export a document as a pdf file.